Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/20/2006
PI Name:	Dulchavsky, Scott A. M.D., Ph.D.		
Project Title:	Advanced Diagnostic Ultrasound in Microgravity		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sdulcha1@hfhs.org	**Fax:**	FY 313 916 9445
PI Organization Type:	PUBLIC SERVICE	**Phone:**	313 916 9306
Organization Name:	Henry Ford Health System		
PI Address 1:	Surgery		
PI Address 2:	2799 W. Grand Boulevard, CFP-1		
PI Web Page:			
City:	Detroit	**State:**	MI
Zip Code:	48202-2608	**Congressional District:**	13
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	ILSRA 2001
Start Date:	05/01/2003	**End Date:**	07/01/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	McCollum, Suzanne	**Contact Phone:**	281 483-7307
Contact Email:	suzanne.g.mccollum@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	In flight development phase (data collection has begun) NOTE: Received NCE to 07/01/2007 per PI (12/06) NOTE: Moved to Henry Ford Hospital from Wayne St U Hosp early 2006 (12/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sargsyan, Ashot (Wyle Laboratories) Hamilton, Douglas (Wyle Laboratories) Melton, Shannon (Wyle Laboratories)		
Grant/Contract No.:	NNJ04HB07A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The diagnosis of acute health problems in space is problematic due to limited training of the Crew Medical Officer (CMO), human and environmental factors, and changes in anatomy and disease presentation in microgravity. Trauma and other acute clinical conditions may cause serious mission ending health consequences. There is no planned radiological capability aboard the ISS further complicating medical diagnosis in space, however, terrestrial investigations have suggested expanded diagnostic applications of ultrasound in abdominal/thoracic trauma, bony and ligamentous injury, dental/sinus infections, opthalmic, and urologic complications. The Health Research Facility (HRF) will have ultrasound capability for research and possibly operational use. This proposal will optimize diagnostic ultrasound in these applications on the ground, confirm the utility in the Microgravity Research Facility, and operationalize the techniques in the ISS to fulfill the objectives of this proposal: Maximize the sensitivity and specificity of ultrasound in the diagnosis of abdominal injury, pneumothorax, extremity injury, dental/sinus infection, opthalmic injury, and urologic disorders in ground based testing. Verify ultrasound in parabolic microgravity using animal models of potential aerospace medical conditions: optimize data compression/transmission algorithms for space medical use. Obtain baseline data from ISS ultrasound flight experiments in normal crew member volunteers to guide CMO inflight use of ultrasound for research/diagnostic requirements on site and through telemedical support. The unique constraints imposed by the space environment require the development of novel strategies for crew member health problems including the expanded use of ultrasound. Thoracic ultrasound, initially investigated by NASA as an alternative diagnostic modality for pneumothorax and orthopedic injury, has proven accuracy in terrestrial and microgravity applications and may have wide spread impact in acute care. The ultrasonic diagnostic modalities described in this proposal, when optimized, would provide a significant, clinically relevant advance in space medical capabilities with profound Earth-based ramifications in rural or military applications. See also: http://www.nasa.gov/
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although some of the techniques investigated in this proposal are appropriate only for a microgravity environment, the majority of the diagnostic and therapeutic algorithms are readily transferable to terrestrial medicine including rural or military applications. The ultrasonic diagnostic modalities described in this proposal, if verified, would provide a significant, clinically relevant advance in space medical capabilities with profound Earth-based ramifications. Furthermore, training and procedural applications in this proposal are designed to be completed by minimally trained, non physicians; verification of this novel concept could significantly extend medical care capabilities in rural or third world situations.
Task Progress:	We have completed the flight data collection component of this proposal and have conducted post-flight debriefs on all crewmembers. Data analysis has been complicated by the proprietary data format of the cardiovascular examinations; we are in the process of obtaining a copy of the NASA software to allow full analysis to commence. We have contributed to a NASA technical brief on Ultrasound use on the ISS for research and medical contingency treatment. We have completed interval analysis of data sets from Increments 8-11 which have resulted in a number of publications on Thoracic, Abdominal, and Musculoskeletal ultrasound performed by the crewmembers. Two publications were written and submitted from the ISS, the only publications with this distinction. The findings of this proposal have demonstrated that non-physician operators can perform diagnostic quality ultrasound examinations with brief, just in time training and remote expert guidance. Preliminary analysis has demonstrated that ultrasound could be used to diagnose medical conditions in the following organ systems: Hepatic, biliary, renal, abdominal viscera, bladder, prostate, cardiovascular, vascular, musculskeletal, dental, sinus, and CNS. We will continue more robust data analysis in the coming year.
Bibliography Type:	Description: (Last Updated: 07/24/2025)
Articles in Peer-reviewed Journals	Fincke EM, Padalka G, Lee D, van Holsbeeck M, Sargsyan AE, Hamilton DR, Martin D, Melton SL, McFarlin K, Dulchavsky SA. "Evaluation of shoulder integrity in space: first report of musculoskeletal US on the International Space Station." Radiology. 2005 Feb;234(2):319-22. Epub 2004 Nov 8. PMID: 15533948 , Feb-2005
Articles in Peer-reviewed Journals	McFarlin K, Sargsyan AE, Melton S, Hamilton DR, Dulchavsky SA. "A surgeon's guide to the universe." Surgery. 2006 May;139(5):587-90. PMID: 16701089 , May-2006
Articles in Peer-reviewed Journals	Chiao L, Sharipov S, Sargsyan AE, Melton S, Hamilton DR, McFarlin K, Dulchavsky SA. "Ocular examination for trauma; clinical ultrasound aboard the International Space Station." J Trauma. 2005 May;58(5):885-9. PMID: 15920397 , May-2005